

Did Working with Women Change the Workplace?

Evidence from Wartime Tokyo *

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Abstract

How do organizations replace workers lost to exogenous exits? We study this question using military conscription during World War II, which removed workers from the Tokyo civil service in a manner that was largely random conditional on occupational role. Using a novel personnel database covering approximately 92,000 unique civil servants from 1928 to 1958, we find that each draft-induced exit increased internal transfers by 0.15 workers and new external hires by 0.22 workers. Offices mitigated disruption by promoting lower-ranked workers and offering bonus salaries to retain remaining staff. The central human resource team facilitated reallocation by reassigning workers from offices with redundant staff, and these donor offices did not increase their own hiring — contradicting theories of vacancy chains. External hires were predominantly male and concentrated in lower-ranked positions, with no effect on the share of female workers. These results reveal that organizations absorb labor supply shocks primarily through their internal labor markets, and that internal reallocation — not external hiring — determines which jobs become available to new entrants.

Keywords: Female representation, Long-term effects, Personnel economics

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1 Introduction

How do workplaces respond to exits of their staff caused by military drafting? Wars mobilize a vast number of workers across sectors through conscription of the active labor force into the military, as well as voluntary exits caused by the expansionary effect of government spending on defense sectors. A vast literature has discussed how displacement accelerated the integration of underrepresented populations into the workforce¹², but the exact adaptation of the labor demand side to military drafting remains untouched. This is staggering, as an employer’s hiring pool should be a joint pool between the internal and external labor market. In particular, military spending often overheats labor markets, and the hiring costs for new workers should induce employers to look for replacements within their organizations or manage the exit by reallocating their existing contracted workers. This shortcoming limits our understanding of (1) the types of jobs that become available to external workers during war periods, and (2) how those jobs are generated through workplaces’ efforts to mitigate disruption by managing their existing human resources.

We document the reactions of workplaces to fill drafting-induced vacancies through recruiting from the internal and external labor market by assembling a novel database on civil servants in wartime Tokyo, which provides the positional information of the universe of its employees. The data allows us to test how managers filled vacancies under their supervision by promoting existing workers, poaching workers from adjacent offices, and hiring from external labor markets. We find that managers’ efforts to fill empty positions varied by the type of job of the drafted workers, and that the worker pool from which managers drew replacements also varied by job type.

The publicly undisclosed drafting process removed workers from the Tokyo civil service, who were distributed across a large hierarchical, multi-layered system. The process and results of the conscription were hidden from the public to prevent eligible workers from avoiding enlistment in the military. Therefore, the distribution of draft-eligible employees across office units within the Tokyo civil service, especially after conditioning on occupational roles, was largely random.

¹The heating of local labor markets further hindered managers’ efforts to find replacements from external labor markets, as government spending to support the war overheated the labor market (?).

²Ferrara (2022) on Black labor supply, Acemoglu et al. (2004); ?; ? on female labor supply

Additionally, the national civil servants were exempt from the drafting, and therefore the vacancy-filling responsibility was largely assigned to the manager of the office unit. As the Tokyo civil service employed a multi-layered hierarchical structure for providing public services to Tokyo's constituents, we identify the impact of drafting by comparing neighboring office units with differential numbers of drafted workers. The service consisted of nearly 12 departments (*kyoku*), each nesting multiple offices (*ka*), and each *ka* including multiple units (*kakari*). Within each *kakari*, employees were further separated into groups by role, ranging from engineering roles to administrative support roles. Office structures across *kakari* units within a *ka* were largely similar to each other, as they shared similar tasks to achieve the goals assigned to their affiliated *ka*. Using the military drafting-induced exits of employees, which the database identifies at the individual level, we estimate how the occupational groups within a *kakari* across its affiliated *ka* changed their composition of workers.

We find that each additional draft-induced exit increases the number of internal transfers into the office by 0.15 workers, driven primarily by transfers from offices in a different section within the same department (0.29 workers) and from different departments (0.15 workers), while within-section transfers are negligible. Each draft also increases new external hires by 0.22 workers. Alongside recruiting workers from internal and external sources, offices mitigated the disruption caused by drafting by promoting workers in lower-ranked positions and offering bonus salaries to the remaining staff to retain them. A one-unit increase in the number of drafted workers increased the probability of a non-drafted coworker being promoted by 8.8 percentage points. Additionally, a similar exit increased the number of existing staff receiving bonus salary (*rinji tokubetsu*) allowances by 0.14 units. We find that retention probability for existing workers was slightly higher in draft-impacted offices (0.9 percentage points per draft), but the estimate is statistically insignificant.

External hires induced by drafting were mainly men, with little effect on the number of women entering the civil service. A one-unit draft-induced exit caused an additional 0.21 male hires, while female hires increased by only 0.01, an estimate that is not statistically significant. As a result, we

do not find any improvement in the share of female workers (0.001, insignificant). Instead, drafting reduced the average length of tenure of staff members in the impacted office unit by 0.04 years per draft.

The results underscore how workplace disruptions due to military drafting were met by workplaces seeking replacements from internal sources rather than the external market. The expertise of workers constitutes a valuable asset for workplaces, and replacing workers with external hires imposes an additional cost (?, ?). The composition of the neighboring organizational structure predicts the success of finding a replacement to fill the role, as the central human resource management team can identify departments with unused employees and relocate them to the impacted office. Such adjustments do not generate any additional hiring at the selected office, as human resources chooses offices in a way that mitigates the overall disruption to the functioning of the workplace, contradicting theories of vacancy chains ().

The responses of workplaces to reallocate workers based on expertise did not create differential exposure to women for incumbent men who were not called up for drafting. The inflow of women into the civil service was mostly associated with voluntary exits of men who appear to have left the civil service to find other working opportunities outside the service. This was fueled by the temporary overheating of the labor market due to increased government spending to support the war. However, the drafting did increase hiring in low-ranked positions, and this may have had lasting effects on the managerial experiences of workers who were promoted as a result of drafting. We find suggestive evidence that managers' wartime exposure to female coworkers is positively associated with the inclusion of women in their offices after the postwar civil service reform, though the instrumental variable estimates are imprecise due to limited statistical power.

2 Background

2.1 Institutional Details of Tokyo Civil Service

The Tokyo public service, comprising the *Tokyo-Fu* and *Tokyo-Shi*, managed public good provision across geographically distinct areas of Tokyo. Although these bodies operated over separate regions, their main offices were centrally located in downtown Tokyo (Figure 1).

In 1944, under intensifying war pressures and facing repeated military defeats, the central government mandated the amalgamation of *Tokyo-Fu* and *Tokyo-Shi* into a single entity, named *Tokyo-To*, through a federal bill. This merger, driven by operational inefficiencies and resource allocation challenges during the war, resulted in significant layoffs, particularly affecting *Tokyo-Shi* employees.

Tokyo-To employed approximately 15,000 highly educated staff, with 30 percent hired through national exams and the remainder locally. The organizational structure was hierarchical, dividing departments based on the public services provided, with some departments split into as many as four layers. Employees were assigned an occupation, such as engineers, managers, clerks, or assistants, along with a classification reflecting their professional expertise and experience. These assignments were determined upon entry into the organization, although transitions across occupations occasionally occurred. The recruitment process for non-assistant employees was merit-based, relying on competitive examinations and internal assessments. Within the civil service, workers were further categorized into upper and lower classes, with managers representing the upper class and clerks or engineers forming the lower class. Leadership positions, such as department heads, were typically filled by personnel recruited from the central government. For many employees, the typical career path involved progressing step by step up the job ladder, with clerks serving as the standard entry point for civil servants aiming to advance. However, the level of competition for promotions among employees was likely influenced by the conscription of a significant portion of the workforce into military service, which temporarily reduced the pool of available candidates. This shift may have also led to substantial changes in both the promotion process and the speed of

advancement for positions such as clerks and managerial staff. In fact, many staff members were conscripted as soldiers, resulting in a severe manpower shortage for managing wartime administration (Tokyo Metropolitan Government 2013).

2.2 Drafting Process of the Japanese Military Service

The Japanese military assembled its forces by calling active service troops and reservists, a process pivotal to our study's analysis of workforce dynamics during and post-conflict periods. The conscription system was designed to ensure a steady supply of personnel by mandating physical examinations for all eligible men upon reaching the age of 20, with those meeting the criteria randomly selected for service.

The drafting process, as outlined in Figure 3, was structured into several stages to efficiently classify conscripts based on their physical capabilities and the military's operational needs. Initially, all eligible men underwent a physical examination to determine their fitness for active duty. Those deemed fit were assigned to active service, where they served as soldiers for a fixed period. Upon completing active duty, these individuals transitioned to the reservist category, ensuring they remained available for military recall during national emergencies. For individuals who did not meet the physical standards for active duty, an alternative route, termed replacement service, was established. These men received training but were not immediately deployed, instead forming a reserve pool for supplementary support when required. Additionally, a subset of candidates was fully exempted from military obligations due to severe health conditions or other qualifying factors.

As the war intensified, particularly in the wake of Japan's challenges during the Second Sino-Japanese War (1937–1945), the drafting system underwent significant modifications to address the military's increasing personnel demands. In 1943, reserve obligations were extended: the period of reserve service following active duty increased from 5–6 years to 15 years, and the secondary reserve period was prolonged until the age of 45. These changes ensured that the military could rely on a broader pool of trained personnel over an extended timeframe, providing critical support for wartime operations.

2.3 The Impact of the War on Tokyo-To

The war significantly altered the workforce composition within *Tokyo-To*, particularly affecting female participation rates and the distribution of staff across various offices. The conscription system, which required eligible men to serve in the military, had profound implications for public institutions, especially in urban centers like Tokyo. Male workers in administrative, engineering, and clerical roles were not exempt from military obligations, leading to their systematic removal from public service positions as they were drafted into active or reserve duty. This mass conscription created substantial disruptions in the operations of public offices and left critical gaps in the workforce that urgently needed to be addressed.

During the war period, the entry of female workers also became a notable feature of *Tokyo-To*'s workforce. Before the war, female employees were largely concentrated in specific departments and were almost absent from the core administrative divisions of the metropolitan government. Their roles were typically limited to auxiliary or supporting tasks, reflecting the gendered labor practices of the time. However, the rapid expansion of labor demand driven by conscription accelerated the entry of women into the workforce. Units were generally structured to include a manager, clerks or engineers, and assistants. Female workers were primarily hired as assistants to support unit leaders in their tasks, often in offices with a lower concentration of engineers. Their continuation and promotion within the organization depended heavily on the approval of their respective unit managers, with their relationships with male incumbents—such as direct subordinates to unit managers or co-workers to clerks and engineers—playing a significant role. In addition to conscription, the demands of wartime administration likely necessitated an increase in the number of staff employed by *Tokyo-To*. The rising responsibilities of the metropolitan government during the war made it essential to expand its workforce. Historical evidence suggest that both the city and the prefecture increased the number of employees and departments during this period (Tokyo Metropolitan Government 2013).

To compensate for the depletion of male workers, *Tokyo-To* rapidly increased the recruitment of women, who were not subject to military service. Women entered the public workforce in un-

precedented numbers, particularly in assistant roles, as highlighted in Figure 5. Panel (a) illustrates the sharp increase in the population of female public civil servants during the World War II (WWII) period, marked by the shaded red area. The count of female employees rose significantly between 1938 and 1944, reflecting a direct response to the conscription of men. However, this trend reversed immediately after the war as male workers returned to their civilian roles, leading to a sharp decline in the number of female public civil servants.

Panel (b) of Figure 5 provides further insight by examining the share of women in the public sector workforce during the same period. The proportion of female public civil servants also increased markedly during WWII, albeit remaining relatively small compared to the overall workforce. The rise in this share highlights not only the increasing reliance on women but also the gendered constraints of wartime employment policies, which often relegated women to lower-ranking and temporary roles within the organizational hierarchy. This unequal distribution persisted even after the war, as men resumed their pre-war positions, and women were often pushed out of public service.

Table 1 shows the difference between offices in 1936 with and without employees that were drafted 2 years later. We find a statistically significant difference between the share of female workers between offices categories. Workers in future drafted offices were more likely to be occupied with male. Since the Japanese military service only conscripted males, the difference in gender balance between offices validates the drafting pattern. The two office categories match each other with respect to its share of new workers and share of higher-ranked workers. We fail to reject similarities between the at least at the 15th confidence level. Drafted and non-drafted employees worked in offices with nearly half of their peers hired at most 4 years before. Also they did not differ in the share of higher-class public servants that they reported to.

3 Data

Our primary data set comes from a historical archive of directory of staff of the Tokyo public servants. The archives include information about the name of the employee, their positions, and their home address. We infer the gender of the staff by using an algorithm trained on historical Japanese names. The personnel record covers from 1928 to 1958.

The directory of staff was published annually with a full list of their employees. The two co-governing institutions (*Tokyo-Shi* and *Tokyo-Fu*) both published separately until the merger of the two institutions in 1943. The Tokyo-Fu directory contains a list of the employees that was drafted into the war. Since we observe the positions of all the employees in a institution, we identify the drafted individual and their surrounding co-workers in a directory.

Our secondary data is the biographical directory of staff at 1942. The biography lists detailed information about higher ranked employees. It includes (1) Date of birth, (2) Prefecture of origin, (3) Education attainment (name of graduating institution) (4) Starting date of entering the public servant system, (5) hobbies, and (6) Physical address. However, the directory was not published annually like the directories, but only published in some years. Also only the Tokyo-Shi and Tokyo-Fu published biographical directories of their employees. In addition the biographical directory only reports the details for employees at least in positions higher than a clerk.

We develop an algorithm that integrates multiple neural networks and computer vision technologies to detect complex layout structures in archival records and extract categorized text data. Given that most modern optical character recognition (OCR) systems are primarily trained on the English language, we utilize a neural network-based OCR developed by the National Diet Library in Japan. This OCR is adept at identifying texts within a document page, which we further analyze using named entity recognition technology to categorize information based on content and layout. We apply this algorithm to a series of historical documents to augment the biographical details of employees. This process generates a comprehensive panel data set of public sector employees in historic Tokyo, detailing their internal assignments by occupation and rank, along with biographic details for a subset of these employees.

Our OCR works better for the recent years than the older years. Since the calligraphy of Japanese letters changed after the war, a modern OCR fails to accurately identify the letters in the directory. Eventhough we use the NDL OCR, which is trained on historical Japanese letters, due to the resolution of the scanned images, the OCR sometimes fail to identify the content. However, we di recover approximately 70% of employees in the historic era, and almost everyone for those in the more modern era (Figure 7).

We predict the gender of the employee by using an external named entity recognition algorithm that predicts the gender of the name. In contrast to Western names, Japanese names tend to reveal the gender of the employee easily. We measure the size and share of female workers of each office unit by aggregating over the pool of employees belonging to a given group. We measure the length of tenure for each employee that we identify in the directory by searching for the first and last year of the directory that we identify their name. Consequently, if an employee is missing in a directory between two directories in which the employee is detected, we treat the employee to be hired in the missing directory. We also infer the location of the office that an employee commuted to by referring to the name of the office unit.

The digitization of the historical archival documents leaves us with a sample of 333,000 observations across 35 years. The data identifies 126,372 unique names and 15,406 female names.

4 Empirical Model

4.1 Drafting specification

We estimate the effect of military drafting on office-level outcomes using three specifications. The unit of observation is position $\times$ year (1938–1945), unless noted. All specifications include year fixed effects and either ka (section) or kyoku (department) fixed effects, plus position fixed effects. Standard errors are clustered at the office level.

Reallocation and hiring (Tables 3, 8, ??). Count outcomes (new hires, transfers in) are mod-

eled with OLS:

$$Y_{j,p,t} = \beta_1 D_{j,p,t} + \beta_2 D_{j,p,t}^{\text{adj}} + \beta_3 \log(M_{j,p,t} + 1) + \gamma + \delta_p + \lambda_t + \varepsilon_{j,p,t}, \quad (1)$$

where $Y_{j,p,t}$ is the number of new hires or transfers into position p in office j in year t ; $D_{j,p,t}$ is the number of male workers drafted from that position; $D_{j,p,t}^{\text{adj}}$ is the number drafted from adjacent positions within the same office; and $M_{j,p,t}$ is the cumulative male stock. The coefficient β_1 measures the change in the expected number of hires or transfers associated with one additional draft from the same position.

Gender composition (Table ??). Female share among new hires or among all workers is modeled by OLS. The sample excludes *kanri* (higher-tier civil servant) positions. Treatment is either an extensive margin indicator (any drafted) or draft share (drafted male count / male baseline). We control for cumulative male baseline and the same fixed effects as above.

Short-run promotion (Table 7). Using a linear probability model, we regress a promotion indicator (rank increased from $t - 1$ to t) on the number of drafts from the same office, for non-drafted workers observed in consecutive years. Fixed effects are year, ka, and position. We interact the treatment with gender, rank, and engineer indicators.

Bonus salary (Table 6). We regress the number of workers receiving *rinji tokubetsu* (temporary special) allowances on the number of drafts, in the 1944 cross-section. The unit of observation is position $\times$ office. Fixed effects are ka and position.

Retention (Table 5). We regress an indicator for whether a non-drafted worker appears in the next year's directory on the number of drafts from the same office. Fixed effects are year, ka, and position.

Postwar tenure (Table 10). For non-drafted workers observed in 1944, we regress postwar tenure (count of years appearing in civil service through 1955) on drafting exposure in their 1944 office. The unit of observation is the individual; treatment is any drafted or draft share; fixed effects are ka (or kyoku) and position as of 1944.

4.2 Identification: conditional independence

Causal inference requires that drafting be as good as random conditional on the fixed effects. Formally, we require that within $ka \times position$ (or $kyoku \times position$) cells and by year, the number of workers drafted from a position is independent of potential outcomes and of other pre-treatment characteristics. This conditional independence assumption is plausible if drafting was determined primarily by national conscription rules (age, health, exemption status) rather than by office-specific factors that also predict hiring or tenure.

Table 2 provides evidence. Panel A shows an unconditional comparison of pre-war (1937–1942) office-year characteristics by whether the office was drafted in 1944. Unconditionally, drafted offices had a higher new-hire share and lower average rank, consistent with drafting being more likely where turnover was higher and workers were lower-tier (and thus less exempt). Panel B conditions on $ka \times modal\ occupation$ and year fixed effects, matching our main specifications. Within these cells, all six pre-war characteristics are uncorrelated with the drafted-in-1944 indicator (p -values 0.31–0.94). Conditional on organizational structure and occupational mix, drafting appears as good as random.

5 Results

5.1 Worker Reallocation in Response to Military Drafting

Table 3 reports OLS estimates of how offices respond to losing male workers to military drafting. The unit of observation is $position \times kakari \times year$ (1938–1945). Column 1 shows that each additional draft in the own position raises total transfers in by approximately 0.15 workers ($\beta = 0.147, p < 0.05$). Table 4 decomposes these transfers by organizational distance: same- ka transfers within the same $kyoku$ are negligible (-0.02 , insignificant), while transfers from a different ka within the same $kyoku$ increase by 0.29 workers per draft ($p < 0.01$) and transfers from a different $kyoku$ increase by 0.15 workers ($p < 0.05$). This indicates that the reallocation response operates

primarily across sections rather than within them. Table 8 reports new hiring. Column 1 shows that each draft raises new hiring by approximately 0.22 workers. All specifications include year, ka, and position fixed effects; standard errors are clustered at the office level.

These results indicate that offices relied on both internal reallocation and external hiring to replace drafted workers, but the two channels operated at different organizational scales. Internal transfers were concentrated at the cross-section level, consistent with a centralized human resource team reassigning workers across ka units within the same department. The near-zero within-ka response suggests that adjacent kakari did not directly exchange workers in response to drafting, and that the reallocation was instead coordinated at a higher organizational level.

5.2 Gender Composition of the Hiring Response

Table 8 investigates whether drafting-induced vacancies are filled differently by gender. Columns 4–6 report OLS estimates for female new hires: the coefficient is small and insignificant (0.007). Columns 7–9 report male new hires: each draft increases male hiring by 0.21 workers. Table ?? reports OLS estimates for female share of the workforce and female share of new hires by rank (R1 = temporary, R2 = regular). The coefficients on the any-drafted indicator are small and statistically insignificant across all four specifications (female share of all workers at R1 and R2, and female share of new hires at R1 and R2). This indicates that male hiring increases in both absolute and proportional terms, while female hiring does not systematically respond to drafting, leaving the gender composition of the workforce and of new hires unchanged.

The absence of a gender composition effect distinguishes the drafting-induced hiring response from the broader wartime trend of rising female employment documented in Section 5.10. While the aggregate female workforce grew substantially during the war, this growth was not driven by the direct replacement of drafted men. Instead, the external labor market supplied predominantly male replacements to drafted offices, and the entry of women into the civil service appears to have been associated with voluntary male exits and the general expansion of wartime administration.

5.3 Short-Run Promotion in Response to Drafting

Table 7 reports OLS estimates of how military drafting affects the promotion probability of non-drafted coworkers. The dependent variable is an indicator equal to one if a worker's rank increased from the prior year (linear probability model). The unit of observation is individual non-drafted workers in consecutive years. Column 1 shows that each additional draft from the same office increases the probability of promotion by 8.8 percentage points ($\beta = 0.088$, $p < 0.05$). Column 2 adds a gender interaction: the female interaction is positive and significant (2.8 percentage points, $p < 0.01$), indicating that women in drafted offices experienced a larger increase in promotion probability than men. Column 3 interacts with rank: the effect is concentrated among rank 2 workers (*shoki/gite*), with no significant differential for rank 1 (*yatoi*) or rank 3 (*shuji/gishi*) positions. Column 4 interacts with an engineer indicator: engineers experience a significantly smaller promotion response (-4.3 percentage points, $p < 0.01$), consistent with engineering roles having more rigid promotion criteria. All specifications include year, ka, and position fixed effects; standard errors are clustered at the office level.

The promotion response reveals an important mechanism through which offices absorbed the loss of drafted workers: rather than relying solely on external replacement, managers advanced existing lower-ranked staff to fill vacated roles. The larger effect for women suggests that drafting may have loosened barriers to female advancement, at least in the short run. The muted response among engineers is consistent with technical positions requiring specialized credentials that limited the scope for internal promotion.

5.4 Bonus Salary Response to Drafting

Table 6 reports OLS estimates of how military drafting affects the number of workers receiving bonus salary (*rinji tokubetsu*) allowances. The unit of observation is position $\times$ office in the 1944 cross-section. Column 1 shows that each additional draft increases the number of workers on bonus salary schedules by 0.14 units ($\beta = 0.136$, $p < 0.05$). Column 2 adds rank interactions: the effect is concentrated among rank 2 workers (0.154, $p < 0.01$), while rank 1 (*yatoi*) workers

show a significantly smaller response ($-0.18, p < 0.01$). Column 3 interacts with an engineer indicator: the bonus response is significantly larger for engineers ($0.15, p < 0.05$), suggesting that offices used compensation to retain workers with specialized skills. Column 4 interacts with the position's female employment share; the interaction is negative but imprecise. All specifications include α and position fixed effects; standard errors are clustered at the office level.

Together with the promotion results, these findings suggest that offices responded to drafting through a combination of internal career advancement and financial incentives. The concentration of bonus allowances among rank 2 workers and engineers indicates that offices targeted retention efforts at the staff most difficult to replace through external hiring, consistent with firm-specific human capital being a key consideration in the replacement strategy.

5.5 Retention of Non-Drafted Workers

Table 5 reports OLS estimates of how military drafting affects the retention of non-drafted coworkers. The dependent variable is an indicator equal to one if the worker appears in the directory in the following year. Column 1 shows that each additional draft is associated with a 0.9 percentage point increase in retention probability, but the estimate is not statistically significant ($p > 0.10$). Column 2 adds a gender interaction: the female interaction is negligible (0.2 percentage points, insignificant). Column 3 adds a rank interaction: the rank 1 interaction is negative (-0.9 percentage points) but insignificant. Column 4 interacts with an engineer indicator: the engineer interaction is similarly small and insignificant. All specifications include year, α , and position fixed effects; standard errors are clustered at the office level.

The lack of a significant retention effect suggests that while offices used promotions and bonus salaries to manage the disruption caused by drafting, these measures did not detectably reduce turnover among remaining staff. This is consistent with retention decisions being driven primarily by factors outside the office's control, such as external labor market conditions and further rounds of conscription.

5.6 Postwar Outcomes of 1944 Colleagues

Table 10 examines postwar outcomes (1946–1955) among non-drafted workers observed in 1944. Columns 1–3 report OLS estimates with z-scored rank change as the dependent variable (postwar average rank minus 1944 rank, standardized within era). The coefficient on the any-drafted indicator (column 1) is 0.058 (approximately 0.06 standard deviations) but imprecise. Column 2 uses draft share instead; the coefficient is 0.073 and similarly imprecise. Column 3 adds a gender interaction: the main effect of draft share remains positive (0.055), the female main effect is negative (−0.57), and the draft share $\times$ female interaction is large and positive (1.06), suggesting that women in more heavily drafted offices may have experienced larger rank gains than men, though the interaction is not statistically significant. Columns 4–5 examine postwar retention (count of years appearing in civil service through 1955): the coefficients on any drafted and draft share are positive (0.037 and 0.050) but imprecise. Columns 6–7 analyze the average female share of workers’ destination office-positions in the postwar period; drafting exposure does not systematically predict the gender composition of postwar placements (coefficients near zero and insignificant). All specifications include α and position fixed effects from 1944; standard errors are clustered at the office level.

The imprecise but consistently positive point estimates for rank advancement and retention suggest that wartime drafting may have modestly accelerated the careers of non-drafted colleagues, though the effects are too noisy to draw firm conclusions. The null result on destination female share indicates that the wartime experience of working alongside women in a drafted office did not systematically alter the gender composition of workers’ postwar placements at the individual level.

5.7 Effect of Drafting on Office Size

Table 11 reports OLS estimates of how military drafting affects remaining office headcount. The unit of observation is $\text{kakari} \times \text{position} \times \text{year}$ (1938–1945). The dependent variable is the number of non-drafted workers remaining in the office. Column 1 shows that each additional male drafted increases remaining headcount by 0.92 workers ($p < 0.01$). Column 2 reveals substan-

tial heterogeneity by rank: the effect is larger for rank 3 (*shuji/gishi*) positions (8.55 additional workers, $p < 0.01$) and significantly smaller for rank 1 (*yatoi*) positions (-2.16 , $p < 0.01$). All specifications include *ka*, position, and year fixed effects; standard errors are clustered at the office level.

The positive coefficient may appear counterintuitive but reflects the fact that offices experiencing drafts were simultaneously receiving transfers and new hires to compensate for the loss. The net effect on remaining headcount thus captures both the mechanical reduction from drafting and the replacement hiring documented in the preceding tables. The large rank 3 effect suggests that offices with drafted senior workers attracted disproportionately more replacements, consistent with greater organizational effort to fill higher-ranked vacancies.

5.8 Characteristics of Donor Offices

Table 12 compares offices that transferred workers out against those that did not, conditional on *ka* $\times$ position fixed effects. The table reports differences in whether the office hired anyone new that year, average tenure length, and female share. These comparisons shed light on whether offices selected to surrender workers had distinct workforce characteristics, offering a test of vacancy-chain theories.

Panel A shows that donor offices are unconditionally more likely to have hired new workers (77% vs. 61%, $p = 0.03$) and have significantly more new hires on average (15.9 vs. 5.3, $p < 0.01$). However, average tenure and female share are indistinguishable between donor and non-donor offices. Panel B conditions on *ka*, position, and year fixed effects: the hiring differential remains marginally significant (12 percentage points, $p = 0.085$), while tenure and female share differences vanish entirely. The conditional hiring difference suggests that offices selected as transfer sources tended to be larger or have higher baseline turnover, but were not systematically different in workforce composition. Importantly, the absence of additional hiring at donor offices conditional on organizational structure is inconsistent with a simple vacancy-chain model, in which poaching from one office should trigger replacement hiring at the source.

5.9 Characteristics of Transferred Workers

Table 14 examines the individual characteristics of workers who were transferred out of donor offices. The regressions test whether transferred workers were disproportionately female, had different tenure lengths, or held different ranks (as a proxy for productivity), conditional on $\text{ka} \times \text{position}$ fixed effects.

Table 15 examines postwar outcomes (1946–1955) for workers who remained in donor offices compared to workers in adjacent *kakari* that did not experience transfers. Workers in donor offices experienced significantly lower postwar rank advancement (-0.31 standard deviations, $p < 0.01$), while postwar tenure was slightly lower but insignificant (-0.24 , $p > 0.10$). The gender interaction in column 2 indicates that women in donor offices experienced an additional tenure penalty (-0.33 years, $p < 0.10$).

The transferred worker characteristics in Table 14 reveal that selection into transfer was not systematically related to gender, tenure, or rank: all three coefficients are small and insignificant. This suggests that the human resource team did not selectively transfer lower-quality workers, but instead reassigned workers based on the organizational needs of the receiving office. The negative postwar rank outcomes for workers remaining in donor offices may reflect the loss of experienced colleagues whose departure reduced opportunities for mentorship or collaborative advancement.

5.10 Women in the Civil Service, 1928–1955

Figure 2 plots the number of women employed in the Tokyo civil service from 1928 to 1955, distinguishing between *kanri* (appointed officials) and *kouri* (public employees). Female participation was effectively zero before 1937 and rose sharply during the war years as offices responded to labor shortages caused by military drafting. The postwar trajectory shows whether wartime gains in female employment persisted through the reform period.

The aggregate pattern reveals that the wartime entry of women into the civil service was not fully reversed after the war. While total female employment declined from its wartime peak, the postwar level remained substantially higher than the pre-war baseline, particularly among *kouri*

employees. This persistence is notable given the return of demobilized soldiers and the broader reversion of wartime labor market conditions, and motivates the analysis in the following subsections of whether individual managers' wartime exposure to female coworkers influenced their subsequent staffing decisions.

5.11 Assignment of Women to Offices Post-Reform

Table 19 reports OLS estimates linking post-reform female inclusion under a manager to the manager's wartime exposure to female coworkers. Columns 1–3 model the probability that a manager includes at least one woman. A 10 percentage point increase in exposure raises this probability by 4.2 points using the maximum exposure measure (column 1) and by 7.2 points using the mean exposure measure (column 2). Column 3 restricts the sample to departmental managers (kacho); the estimate is positive but imprecise. Columns 4–6 report the extensive margin: a 10 percentage point increase in exposure is associated with 0.1–0.2 additional women in the manager's office unit, depending on the measure. The first-stage relationship and IV results are discussed in the next subsection.

5.12 The Causal Effect of Exposure on Post-Reform Female Assignment

Table 20 compares OLS and IV estimates of how wartime exposure to female coworkers affects the post-reform assignment of women to offices. The OLS coefficient is positive and significant, consistent with Table 19. The IV estimate (Panel A, column 2) points in the same direction and is larger in magnitude (5.03 vs. 1.73) but only marginally significant ($p < 0.10$), with a weak first-stage F-statistic (3.19). Columns 3–4 restrict to managers with wartime exposure; the IV estimate remains positive but the first-stage strength collapses ($F = 0.07$). The results do not allow rejection of a zero causal effect; interpretation must emphasize limited statistical power rather than a precise null.

5.13 Network Proximity and Transfer Selection

To investigate whether organizational reallocation follows existing workplace networks, we construct a measure of past co-worker connections between donor and destination offices. For each worker observed in year $t-1$, we check whether any colleague in their current office previously shared a workplace (office $\times$ ka) with anyone currently working at an office that experienced military drafting in year t . We define “self-connected” workers as those who themselves have such a past link. Table 16 reports the results.

Panel A of Table 16 reports the informativeness of this network measure. We document the share of workers for whom the connection measure is non-zero, which is critical for interpreting the regression results: if the measure is zero for nearly all workers, we cannot distinguish between null effects and an underpowered measure. Panel B reports OLS estimates of whether past connections predict transfer probability, conditional on year, ka, and position fixed effects. Column 1 tests whether working in an office with any past connection to a drafted destination predicts being transferred. Column 2 tests the individual-level measure: whether a worker who themselves previously co-worked with destination office members is more likely to be transferred. Column 3 includes both measures simultaneously to distinguish office-level from individual-level selection. Column 4 uses the intensive margin (number of connected colleagues). Column 5 interacts the self-connection measure with gender.

These results shed light on the mechanism underlying the reallocation response documented in Table 3. If transfers are coordinated through personal networks—as in referral-based hiring—we would expect positive coefficients on the connection measures. If instead transfers are centrally assigned by the personnel office without regard to prior workplace ties, the coefficients should be small and insignificant.

5.14 Long-Run Outcomes of Transferred Workers

Table 17 compares the long-run career outcomes of transferred workers against two natural reference groups: peers at the destination office (Panel A) and peers who remained at the donor office

(Panel B). The dependent variables capture tenure after transfer, postwar survival in the civil service (appearing in any year 1947–1955), number of postwar years observed, and rank advancement (maximum rank achieved minus rank at transfer).

Panel A conditions on destination office $\times$ transfer year fixed effects, so the comparison isolates whether transferred workers fare differently from the incumbents they joined. Panel B conditions on donor office $\times$ transfer year fixed effects, comparing transferred workers to the colleagues they left behind. The key question is whether wartime transfers conferred a career advantage or disadvantage: did the disruption of being moved accelerate or hinder career progression relative to staying in place?

5.15 Post-War Retention of Wartime Hires

Table 18 examines whether workers hired during the wartime labor shortage persisted in the civil service after the war ended. We identify 21,410 wartime new hires (workers whose first appearance falls in 1940–1945), of whom 1,993 were female and 19,417 male. The overall postwar retention rate—defined as appearing in any year between 1947 and 1955—is 15.6%. Strikingly, the female retention rate is substantially lower at 7.6%, compared to 16.5% for males.

Panel A reports the first stage: the relationship between office-level draft intensity and new hiring. The coefficient on the number of males drafted is positive but imprecise (0.11, SE 0.21), indicating that draft intensity is a weak instrument for hiring at the office level. This reflects the fact that hiring decisions are driven by many factors beyond the immediate draft vacancy, including administrative expansion and budget allocation.

Panel B reports reduced-form estimates of draft intensity on postwar outcomes. While the share of wartime hires retained or promoted shows no significant relationship with draft intensity, the number of workers promoted postwar is significantly *lower* in offices with greater wartime drafting (-0.064 , $p < 0.05$). This suggests that workers hired into heavily drafted offices may have been negatively selected or that these offices experienced greater postwar reorganization.

For the gender-specific analysis, we restrict the female sample to low-rank administrative po-

sitions (rank 1–2), as these are the positions most sensitive to the drafting shock. The first stage for female low-rank hires is also weak (coefficient 0.024, SE 0.034), reflecting the small number of female hires in any given office. The reduced-form retention estimate for women is positive (0.007) but imprecise. The IV estimates, presented in Panel C, are correspondingly noisy due to the weak first stage and should be interpreted with caution.

5.16 Comparison with Related Literature

Our findings contribute to and extend several strands of the literature on worker substitutability, wartime labor market disruptions, and the persistence of female employment gains.

Worker substitutability and replacement costs. Jäger et al. (2025) analyze 34,000 unexpected worker deaths in Germany and find that coworker earnings increase by approximately 0.6% in the year following the death, with cumulative effects of roughly 6,000 EUR (18% of the deceased worker’s annual earnings) over five years. The positive wage effects are concentrated among same-occupation coworkers, while workers in different occupations experience wage *decreases* when a high-skilled worker or manager dies. Our context differs in that the exiting workers are drafted rather than deceased, and the organization responds through a combination of internal transfers, new hiring, and internal promotion rather than solely through wage adjustments. Our finding that offices respond to drafts with an 8.8 percentage point increase in promotion probability for remaining workers is consistent with Jäger et al.’s evidence that firms use internal labor market mechanisms to adjust to worker exits. However, while Jäger et al. find positive wage effects and increased retention, we find null retention effects and instead document a promotion-and-bonus response. This difference likely reflects the institutional context: in the rigid salary schedules of the Japanese civil service, promotions and bonus allowances (*rinji tokubetsu*) served the role that wage adjustments play in private German firms.

Both studies confirm that worker exits create substantial adjustment costs and that incumbents and outsiders are imperfect substitutes. Jäger et al. find that these effects are larger in thin external

labor markets; analogously, our finding that offices rely heavily on cross-section internal transfers (Table 4) rather than external hiring suggests that the wartime civil service operated in a thin market where replacement costs were high.

Female labor supply and wartime mobilization. Acemoglu et al. (2004) exploit cross-state variation in male mobilization rates during World War II and find that a 10 percent increase in female labor supply reduced female wages by 7–11% and male wages by 4–6%. At midcentury, women were closer substitutes for high-school-educated men than for those with lower or higher skills. Goldin and Olivetti (2013) demonstrate that the wartime impact on female labor supply persisted through 1950 and 1960, particularly among women in the top half of the education distribution.

Our setting provides a complementary perspective from a non-Western bureaucratic context. Unlike the U.S., where wartime female labor supply increases were substantial and persistent, our data show that the *direct* replacement of drafted men was predominantly male—female hiring did not respond to office-level drafting (Table 8). The aggregate rise in female civil service employment documented in Figure 2 was thus not driven by the direct replacement channel but rather by the general expansion of wartime administration and voluntary male exits. This finding challenges the standard narrative, derived primarily from U.S. evidence, that wartime labor shortages mechanically drew women into male-dominated occupations.

Nevertheless, consistent with Acemoglu et al. (2004) and Goldin and Olivetti (2013), the wartime gains in female employment were not fully reversed. Our postwar retention analysis shows that 7.6% of wartime female hires remained in the civil service through 1947–1955, and the aggregate time series confirms persistently higher female representation. The lower female retention rate compared to males (7.6% vs. 16.5%) is consistent with the pattern documented by Rose (2018), who shows that much of the wartime female labor force participation in the U.S. was reversed by 1950, with persistence concentrated among older married women.

Wartime wage data. Unlike the U.S. context studied by Goldin and Margo (1992) and Acemoglu et al. (2004), where detailed wage data allow estimation of the wage effects of female labor supply shocks, the Japanese civil service operated under centralized salary schedules that limit within-position wage variation. Our data record salary levels but not the detailed pay grades within positions. Japan’s Basic Survey on Wage Structure (*Chingin Kōzō Kihon Tōkei Chōsa*) began in 1948 and reports gender-disaggregated wages from that year onward, but historical data from the wartime period covering public sector employees by gender are not available in published form. The female-to-male wage ratio in Japan was approximately 60% in the late 1980s and has risen to about 72% by 2014, but comparable figures for the 1940s civil service cannot be constructed from available sources. Our analysis of bonus salary (*rinji*) allowances in Table 6 provides the closest available proxy for compensation responses to labor shortages.

Persistence and the transmission of gender norms. Aneja et al. (2024) study how the expansion of female clerical employment in the U.S. federal government during World War I persistently shifted gender norms, finding intergenerational effects on daughters’ labor supply and earnings. Our manager-exposure results (Tables 19 and 20) document a related but distinct channel: managers who worked alongside women during the war were more likely to integrate women into their postwar offices. While our IV estimates lack power due to a weak first stage, the consistent positive OLS relationship suggests that direct workplace exposure may have shifted managers’ willingness to hire women, complementing the intergenerational transmission channel documented by Aneja, Farina, and Xu.

6 Conclusion

This paper examines how organizations respond to exogenous labor supply shocks and how these responses shape long-run workforce composition, leveraging a unique historical shock: the mass conscription of male workers during World War II in the Tokyo civil service. Using a novel personnel dataset covering 92,000 workers from 1928 to 1958, we trace the full chain of organizational

responses—from immediate reallocation through postwar persistence.

Our findings reveal several key insights. First, offices responded to drafting through a combination of internal transfers (0.15 additional workers per draft, concentrated at the cross-section level), new hiring (predominantly male), and internal promotion of remaining workers (8.8 percentage points per draft). The reliance on cross-section rather than within-team transfers, combined with the use of bonus salary allowances to retain scarce workers, points to a centralized human resource function managing the reallocation.

Second, the direct replacement of drafted men was overwhelmingly male: female hiring did not respond to office-level drafting, and the gender composition of affected offices remained unchanged. The aggregate wartime increase in female civil service employment was instead associated with the general expansion of wartime administration rather than the mechanical replacement of drafted workers. This finding challenges the standard narrative from the U.S. literature that wartime labor shortages directly drew women into male-dominated occupations.

Third, we examine the long-run consequences of wartime disruptions. Wartime hires had a postwar retention rate of only 15.6%, with women retained at roughly half the male rate (7.6% vs. 16.5%). At the manager level, wartime exposure to female coworkers predicted greater postwar female integration, though IV estimates lack statistical power. The persistence of female employment gains despite low individual retention rates suggests that the wartime experience normalized female participation at the organizational level.

By documenting the complete organizational response to labor supply shocks—from immediate replacement through long-run persistence—this study contributes to the literatures on worker substitutability (Jäger et al., 2025), wartime female labor supply (Acemoglu et al., 2004; Goldin and Olivetti, 2013), and the transmission of gender norms (Aneja et al., 2024). Our results highlight that the mechanisms of persistence operate differently in a bureaucratic context with centralized personnel management than in the decentralized private-sector labor markets studied in prior work.

Table 1: Summary Statistics

Panel A: Sample Counts

	Full Sample	Drafted Years	1944
Unique workers	92,439	54,254	9,323
Unique kyoku	119	48	8
Unique ka (ka $\times$ kyoku)	4,967	1,464	276
Unique kakari (kakari $\times$ ka $\times$ kyoku)	9,234	1,690	383
Unique women	5,968	3,720	702
Workers at rank 3 (shiji/gishi)	15,296	4,989	470
Workers at rank 2 (shoki/gite)	124,127	74,352	8,013
Workers at rank 1 (yatoi/shokutaku)	99,555	29,312	789

Panel B: Kakari-Level Averages

	Full Sample	Drafted Years	1944
Workers per kakari	22.7 (102.4)	58.3 (206.6)	24.3 (38.7)
Unique positions per kakari	2.2 (1.7)	3.4 (2.6)	3.4 (2.3)
Workers per rank-3 role	3.5 (22.5)	5.3 (47.2)	2.3 (3.8)
Workers per rank-2 role	12.8 (48.5)	19.3 (66.1)	8.5 (14.7)
Workers per rank-1 role	11.5 (36.1)	23.6 (67.4)	6.3 (8.3)

Notes: Panel A reports unique counts for the full sample, the subset of years with any drafted workers (1938, 1940, 1941, 1943, 1944), and 1944 alone. Rank 3 = *shiji/gishi* (supervisory); Rank 2 = *shoki/gite* (clerical/technical staff); Rank 1 = *yatoi/shokutaku* (temporary/contract). Panel B reports kakari-level averages (standard deviations in parentheses). A kakari is the lowest organizational unit (team), nested within ka (section) and kyoku (bureau).

Table 2: Balance Test: Pre-War Characteristics (1937–1942) by Future Drafting Status (1944)

Panel A: Unconditional Comparison

	Drafted in 1944		Not Drafted		Diff	<i>p</i> -value
	Mean	SD	Mean	SD		
Female Share	0.027	0.106	0.172	0.305	-0.145	<0.001
New Hire Share	0.335	0.366	0.152	0.258	0.183	<0.001
Promoted Share	0.021	0.105	0.017	0.093	0.004	0.754
Ext. Transfer Share	0.217	0.308	0.228	0.343	-0.011	0.744
N Workers	38.682	120.591	27.200	72.530	11.482	0.163
Avg Rank	2.154	0.833	2.052	0.577	0.102	0.099
Offices	28		21			
Obs (kakari $\times$ pos $\times$ year)	512		135			

Panel B: Conditional Comparison (Ka + Occupation + Year FE)

	Coef	SE	<i>t</i> -stat	<i>p</i> -value	<i>N</i>
Female Share	0.0018	0.0166	0.11	0.916	626
New Hire Share	—	—	—	—	—
Promoted Share	-0.0251	0.0232	-1.08	0.287	319
Ext. Transfer Share	-0.0843	0.2199	-0.38	0.703	626
N Workers	9.7028	16.7093	0.58	0.564	626
Avg Rank	-0.0837	0.0675	-1.24	0.221	626

Notes: Sample restricted to offices observable in both the pre-war period (1937–1942) and in 1944. Unit of observation: kakari $\times$ position $\times$ year, matching the regression panel. Panel A reports unconditional means split by whether the office experienced any military drafting in 1944. Panel B reports coefficients from OLS regressions of each characteristic on a drafted-in-1944 indicator, with ka, occupation, and year fixed effects. Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Worker Reallocation: Transfers In

Model:	Baseline (1)	× Rank (2)	× Eng. (3)	× Gender (4)
<i>Variables</i>				
No. drafted workers	0.1466** (0.0599)	0.1341*** (0.0359)	0.2079 (0.1440)	0.1741*** (0.0619)
No. drafted × Rank 1		0.1538 (0.2452)		
No. drafted × Rank 3		0.8616** (0.4210)		
No. drafted × Engineer			-0.0865 (0.1356)	
<i>Fixed-effects</i>				
year_num	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Dependent variable: number of workers who transferred in from a different bureau group. Transfers are defined as workers whose normalized bureau (kyoku) group changed from the prior year, using a crosswalk that accounts for the 1943 Tokyo-Fu/Shi to Tokyo-To merger. Column (2) interacts the treatment with rank indicators (rank 1 = *yatoi/shokutaku*; rank 3 = *shuji/gishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Column (4) interacts with the position’s female employment share. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Transfer Decomposition by Organizational Distance

Model:	Same ka & kyoku			Same kyoku, diff. ka			Different kyoku		
	Base (1)	× Rank (2)	× Eng. (3)	Base (4)	× Rank (5)	× Eng. (6)	Base (7)	× Rank (8)	× Eng. (9)
<i>Variables</i>									
No. drafted workers	-0.0205 (0.0148)	-0.0134 (0.0156)	-0.0419 (0.0281)	0.2910*** (0.0838)	0.3462*** (0.0722)	0.1492 (0.1336)	0.1466** (0.0599)	0.1341*** (0.0359)	0.2079 (0.1440)
No. drafted × Rank 1		-0.0722 (0.0497)			-0.6017*** (0.1517)			0.1538 (0.2452)	
No. drafted × Rank 3		0.4560* (0.2421)			1.047** (0.5159)			0.8616** (0.4210)	
No. drafted × Engineer			0.0302 (0.0345)			0.1998 (0.1372)			-0.0865 (0.1356)
<i>Fixed-effects</i>									
year_num	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>									
Observations	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Dependent variable: number of workers arriving from each organizational distance. Columns (1)–(3): workers who transferred from a different kakari within the same ka and kyoku (bureau) group. Columns (4)–(6): workers from the same kyoku group but a different ka (section). Columns (7)–(9): workers from a different kyoku group entirely. Kyoku groups use a crosswalk that accounts for the 1943 Tokyo-Fu/Shi to Tokyo-To merger. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: Retention of Non-Drafted Workers

<i>Dep. var: Retained (=1 if worker appears in year t)</i>				
Model:	Baseline (1)	$\times$ Gender (2)	$\times$ Rank (3)	$\times$ Engineer (4)
<i>Variables</i>				
No. drafted workers	0.0091 (0.0062)	0.0090 (0.0058)	0.0129 (0.0098)	0.0128 (0.0085)
No. drafted $\times$ Female		0.0021 (0.0139)		
No. drafted $\times$ Rank 1			-0.0089 (0.0091)	
No. drafted $\times$ Engineer				-0.0119 (0.0090)
Female		-0.0080 (0.0060)		
<i>Fixed-effects</i>				
year_num	Yes	Yes	Yes	Yes
Ka	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	190,187	190,187	190,187	190,187

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS linear probability model. Dependent variable: retained (=1 if worker appears in year t). Treatment: number of male workers drafted from the same office $\times$ position in year t . All four columns use the full sample of non-drafted workers observed in any year from 1937 to $t - 1$, using their most recent observation for office and position assignment. Column 1 estimates the baseline effect. Column 2 interacts the treatment with a female indicator. Column 3 interacts with rank dummies (Rank 1 = low-rank/temporary; Rank 3 = high-rank; Rank 2 is the omitted category). Column 4 interacts with an engineer-position indicator. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 6: Bonus Salary (*Rinji Tokubetsu*) Response to Drafting
Dep. var: No. workers receiving rinji tokubetsu

Model:	Baseline (1)	× Rank (2)	× Engineer (3)	× Gender (4)
<i>Variables</i>				
No. drafted × Female share				-0.4492 (0.3928)
No. drafted workers	0.1361** (0.0522)	0.1538*** (0.0578)	0.0230 (0.0430)	0.1569** (0.0696)
No. drafted × Rank 1		-0.1806*** (0.0614)		
No. drafted × Rank 3		0.2938 (0.1793)		
No. drafted × Engineer			0.1481** (0.0744)	
Female share				1.854* (0.9597)
<i>Fixed-effects</i>				
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	576	576	576	576

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regression using 1944 cross-section. Unit: position × office. Dependent variable: number of non-drafted workers receiving *rinji tokubetsu* (temporary special allowance). Treatment: number of male workers drafted from the same office × position. Column (1) is the baseline specification. Column (2) interacts the treatment with rank indicators (Rank 2 = regular civil servant is the base; Rank 1 = *yatoi/shokutaku*; Rank 3 = *shuji/gishi*). Column (3) interacts with an engineer indicator. Column (4) interacts with the female share of workers in the position. All specifications include ka and position fixed effects and control for log cumulative male baseline. Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 7: Short-Run Promotion in Response to Drafting

<i>Dep. var: Promoted (=1 if rank increased from $t-1$ to t)</i>				
Model:	Baseline (1)	$\times$ Gender (2)	$\times$ Rank (3)	$\times$ Engineer (4)
<i>Variables</i>				
No. drafted workers (office)	0.0881** (0.0400)	0.0891** (0.0399)	0.0820** (0.0356)	0.0923** (0.0381)
No. drafted $\times$ Female		0.0279*** (0.0079)		
Female		-0.0220 (0.0183)		
No. drafted $\times$ Rank 1			0.0353 (0.0516)	
No. drafted $\times$ Rank 3			-0.0240 (0.0268)	
No. drafted $\times$ Engineer				-0.0425*** (0.0111)
<i>Fixed-effects</i>				
year_num	Yes	Yes	Yes	Yes
Ka	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	8,754	8,754	8,754	8,754

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions (linear probability model). Dependent variable: promoted (=1 if worker's position rank increased from year $t-1$ to t). Sample: non-drafted workers present in the same office in consecutive years. Treatment: total number of male workers drafted from the same office in year t (office-level variation). Column 2 interacts treatment with a female indicator; Column 3 interacts with rank indicators (rank 2 is the omitted category; rank 1 = temporary *yatoi/shokutaku*, rank 3 = supervisory *shiji/gishi*); Column 4 interacts with an engineer indicator (positions containing *gi/engineer*). Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 8: New Hiring in Response to Military Drafting

Model:	No. new hires			No. new female			No. new male		
	Baseline (1)	× Rank (2)	× Eng. (3)	All (4)	Rank 1 (5)	Rank 2+ (6)	All (7)	Rank 1 (8)	Rank 2+ (9)
<i>Variables</i>									
No. drafted workers	0.2171** (0.1091)	0.2513** (0.1042)	0.0959 (0.0869)	0.0073 (0.0061)	-0.0047 (0.0042)	0.0120** (0.0055)	0.2126 (0.1372)	-0.0680 (0.0420)	0.2806** (0.1104)
No. drafted × Rank 1		-0.3642** (0.1444)							
No. drafted × Rank 3		1.129** (0.5364)							
No. drafted × Engineer			0.1709 (0.1486)						
<i>Fixed-effects</i>									
year_num	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>									
Observations	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses

Signif. Codes: ***: 0.01, **: 0.05, *: 0.1

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Columns (1)–(3): dependent variable is total new hires. Columns (4)–(6): new female hires (all, rank 1, rank 2+). Columns (7)–(9): new male hires (all, rank 1, rank 2+). Column (2) interacts the treatment with rank indicators (rank 1 = *yatoi/shokutaku*; rank 3 = *shuji/gishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 9: Gender Composition and Tenure

Dep. var:	Female share			Avg. tenure		
Model:	Baseline	× Rank	× Eng.	Baseline	× Rank	× Eng.
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Variables</i>						
No. drafted workers	0.0006 (0.0009)	0.0005 (0.0009)	0.0026 (0.0021)	-0.0373*** (0.0116)	-0.0275** (0.0114)	-0.0793*** (0.0216)
No. drafted × Rank 1		0.0015 (0.0026)			-0.0800* (0.0470)	
No. drafted × Rank 3		-0.0122 (0.0132)			-0.2095 (0.1490)	
No. drafted × Engineer			-0.0029 (0.0022)			0.0614** (0.0244)
<i>Fixed-effects</i>						
year_num	Yes	Yes	Yes	Yes	Yes	Yes
Ka	Yes	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	6,593	6,593	6,593	6,593	6,593	6,593

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions at the position × year level (1938–1945). Columns (1)–(3): dependent variable is female share of all workers. Columns (4)–(6): dependent variable is average tenure (years in data). Column (1) and (4) report baseline effects. Columns (2) and (5) interact draft exposure with rank indicators (rank 1 = *yatoi*/temporary; rank 3+ = senior; rank 2 = base category, absorbed by position FE). Columns (3) and (6) interact draft exposure with an engineer indicator (positions containing *gi/engineer*). All specifications include $\log(\text{cumulative male baseline} + 1)$ as a control (not displayed), as well as year, ka, and position fixed effects. Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 10: Postwar Outcomes of 1944 Colleagues (1946–1955)

Dependent Variables:	rank_z_change Promotion		avg_female_share_dest Dest. F.Share	
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Draft share	-0.0919 (0.1438)	-0.0929 (0.1457)	-0.0018 (0.0070)	-0.0018 (0.0059)
Female (=1)		-0.2087 (0.1482)		0.1283*** (0.0477)
Draft share $\times$ Female		0.0047 (0.7866)		0.0310 (0.1611)
<i>Fixed-effects</i>				
ka_id_1944	Yes	Yes	Yes	Yes
pos_norm_1944	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	1,281	1,281	1,281	1,281
R ²	0.46085	0.46163	0.09966	0.24099

Clustered (office_id_1944) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Columns (1)–(2): dependent variable is z-scored rank change (postwar avg. minus 1944 rank, standardized within era). Column (2) includes gender interaction. Columns (3)–(4): dependent variable is average female share of the worker's office-position across postwar years. Column (4) includes gender interaction. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock. Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Effect of Military Drafting on Remaining Office Size

Model:	Baseline (1)	× Rank (2)	× Eng. (3)	× Gender (4)
<i>Variables</i>				
No. drafted workers	0.9225*** (0.2762)	1.120*** (0.3167)	1.385* (0.7122)	0.7134*** (0.2369)
No. drafted × Rank 1		-2.159*** (0.6828)		
No. drafted × Rank 3		8.553*** (3.112)		
No. drafted × Engineer			-0.6566 (0.6895)	
<i>Fixed-effects</i>				
year_num	Yes	Yes	Yes	Yes
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	7,409	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Unit of observation: position × kakari × year (1938–1945). Dependent variable: remaining headcount (total workers minus drafted workers). Column (2) interacts the treatment with rank indicators (rank 1 = *yatoilshokutaku*; rank 3 = *shujilgishi*; rank 2 is the omitted category). Column (3) interacts with an engineer indicator. Column (4) interacts with the position’s female employment share. Rank and engineer main effects are absorbed by position fixed effects. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline (not shown). Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 12: Characteristics of Donor Offices (Offices with Transfers Out)

Panel A: Unconditional Comparison

	Donor		Non-Donor			
	Mean	SD	Mean	SD	Diff	<i>p</i> -value
Has New Hire (=1)	0.729	0.445	0.552	0.497	0.177	<0.001
No. New Hires	9.251	18.899	3.317	9.719	5.934	<0.001
Avg Tenure (years)	2.880	1.301	2.636	1.516	0.245	<0.001
Female Share	0.026	0.079	0.044	0.142	-0.019	<0.001

Panel B: Conditional Comparison (Kyoku + Position + Year FE)

	Coef	SE	<i>t</i> -stat	<i>p</i> -value	<i>N</i>
Has New Hire (=1)	0.0774**	0.0343	2.26	0.024	6,603
No. New Hires	3.9355**	1.7100	2.30	0.022	6,603
Avg Tenure (years)	0.1218***	0.0364	3.34	<0.001	6,603
Female Share	-0.0079**	0.0035	-2.29	0.022	6,603

Notes: Sample: $ka \times position \times year$ (1938–1945). A *donor* office is one that had at least one worker transfer out in that year. Panel A reports unconditional means. Panel B reports coefficients from OLS regressions of each characteristic on a donor indicator, with *kyoku*, position, and year fixed effects, controlling for log office headcount. Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 13: Lagged Hiring Test: Prior-Year Outcomes by Future Donor Status

Panel A: Unconditional Comparison

	Future Donor		Non-Donor		Diff	<i>p</i> -value
	Mean	SD	Mean	SD		
Has New Hire (=1)	0.696	0.460	0.478	0.500	0.218	<0.001
No. New Hires	7.447	17.243	1.867	4.813	5.580	<0.001
Avg Tenure (years)	2.724	1.294	3.237	1.908	-0.513	<0.001
Female Share	0.034	0.098	0.046	0.158	-0.012	<0.001

Panel B: Conditional Comparison (Kyoku + Position + Year FE + Log Headcount at *t*)

	Coef	SE	<i>t</i> -stat	<i>p</i> -value	<i>N</i>
Has New Hire (=1)	0.1603***	0.0156	10.29	<0.001	7,235
No. New Hires	5.1419***	0.6665	7.71	<0.001	7,235
Avg Tenure (years)	0.1133***	0.0289	3.92	<0.001	7,235
Female Share	-0.0081**	0.0032	-2.53	0.012	7,235

Notes: This table tests whether the new-hiring gap between donor and non-donor offices reflects vacancy chains or selection. Outcomes (new hires, tenure, female share) are measured at year $t-1$; donor status is defined at year t (whether the office had any transfers out). If offices that will become donors already exhibit higher hiring rates in the prior year, this suggests selection rather than vacancy chains. Sample: $ka \times$ position, outcome years 1937–1944, donor status years 1938–1945. Panel A reports unconditional means. Panel B reports coefficients from OLS regressions with kyoku, position, and year fixed effects, controlling for log office headcount at year t . Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 14: Characteristics of Transferred Workers

Model:	Female (1)	Tenure (2)	Rank (3)	All (4)
<i>Variables</i>				
Female (=1)	0.0196 (0.0142)			0.0200 (0.0136)
Tenure (years)		0.0008 (0.0011)		0.0012 (0.0011)
Rank (enriched)			-0.0104 (0.0114)	-0.0112 (0.0114)
<i>Fixed-effects</i>				
year_num	Yes	Yes	Yes	Yes
lag_ka_id	Yes	Yes	Yes	Yes
lag_pos	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	49,036	49,036	49,036	49,036

Clustered (lag_office) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions (linear probability model). Dependent variable: transferred (=1 if worker moved to a different unit from prior year). Unit of observation: individual worker $\times$ year (1938–1945), restricted to workers with prior-year records. Rank is an enriched 1–5 scale serving as a productivity proxy. All specifications include year, ka (prior-year), and position (prior-year) fixed effects. Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 15: Postwar Outcomes of Remaining Workers in Donor vs. Non-Donor Offices

Model:	Tenure		Promotion	
	(1)	(2)	(3)	(4)
<i>Variables</i>				
Donor office (=1)	-0.2414 (0.2218)	-0.2117 (0.2308)	-0.3109*** (0.0837)	-0.3179*** (0.0837)
Female (=1)		-0.5761*** (0.0927)		-0.2180* (0.1217)
Donor $\times$ Female		-0.3260* (0.1917)		
<i>Fixed-effects</i>				
ka_id	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	3,385	3,385	1,276	1,276
R ²	0.09043	0.09689	0.42777	0.42871

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Columns (1)–(2): dependent variable is postwar tenure (count of years appearing 1946–1955). Columns (3)–(4): dependent variable is z-scored rank change (postwar avg. minus 1944 rank). A *donor* office had at least one worker transfer out during 1938–1945. The comparison group is same-ka offices (kakari $\times$ position) that did not have transfers out. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock and kakari $\times$ position headcount (not shown). Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 16: Network Proximity and Transfer Probability

	Office	Self	Both	Intensive	× Gender
<i>Dep. var: Transferred (=1 if worker moved to different ka/kyoku in year t)</i>					
Office has connection to drafted dest.					
Self connected to dest. office					
No. connected colleagues					
<i>Panel A: Informativeness of Network Measure</i>					
Share with any connection					[Computation in progress]
Share self-connected					[Computation in progress]
Mean connections					[Computation in progress]
Year + Ka + Position FE	Yes	Yes	Yes	Yes	Yes

Notes: OLS linear probability model. Dependent variable: transferred (=1 if worker moved to a different ka or kyoku in year t). Sample: all workers observed in year $t-1$ (1938–1945). “Office has connection” = 1 if any colleague in the worker’s office at $t-1$ previously shared a workplace (office × ka) with anyone currently at an office that experienced military drafts in year t . “Self connected” = 1 if the worker themselves had such a past workplace link. Panel A reports the share of workers for whom each measure is non-zero. All specifications include year, ka, and position fixed effects. Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Results pending completion of network computation.

Table 17: Long-Run Outcomes of Transferred Workers

Panel A: Transferred vs. Destination Office Peers

Model:	Tenure After (1)	Postwar Surv. (2)	Postwar Yrs (3)	Rank Gain (4)
<i>Variables</i>				
Transferred (=1)	0.6888*** (0.0388)	0.0436*** (0.0029)	0.1325*** (0.0093)	0.0467*** (0.0034)
<i>Fixed-effects</i>				
Dest. Kakari	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Position	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,198,457	5,198,457	5,198,457	5,198,457
<i>Clustered (Dest. Kakari) standard-errors in parentheses</i>				
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>				

Panel B: Transferred vs. Donor Office Peers

Model:	Tenure After (1)	Postwar Surv. (2)	Postwar Yrs (3)	Rank Gain (4)
<i>Variables</i>				
Transferred (=1)	1.254*** (0.0428)	0.0591*** (0.0025)	0.1700*** (0.0088)	0.0559*** (0.0028)
<i>Fixed-effects</i>				
Donor Kakari	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
Position	Yes	Yes	Yes	Yes
<i>Fit statistics</i>				
Observations	5,766,663	5,766,663	5,766,663	5,766,663
<i>Clustered (Donor Kakari) standard-errors in parentheses</i>				
<i>Signif. Codes: ***: 0.01, **: 0.05, *: 0.1</i>				

Notes: OLS regressions comparing transferred workers to peers. Panel A compares transferred workers to non-transferred workers already at the destination office in the transfer year; Panel B compares to workers who remained at the donor office. “Tenure After” = years observed after the transfer year. “Postwar Surv.” = 1 if worker appears in any year 1947–1955. “Postwar Yrs” = number of years observed 1947–1955. “Rank Gain” = maximum rank achieved minus rank at time of transfer. Panel A includes destination kakari, year, and position FE; Panel B includes donor kakari, year, and position FE. Kakari is the minimum organizational unit (section) within each office. Standard errors clustered at the office-ka level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 18: Post-War Retention of Wartime Hires: IV Estimates

Panel A: First Stage (Dep. var: No. new hires)

Model:	All Hires (1)	Male Hires (2)	Female (Low-Rank) (3)
<i>Variables</i>			
No. males drafted	0.1065 (0.2094)	0.0820 (0.1952)	0.0244 (0.0340)
<i>Fixed-effects</i>			
year_num	Yes	Yes	Yes
norm_kyoku	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	1,318	1,318	1,318
Wald (joint nullity)	53.330	48.303	20.023

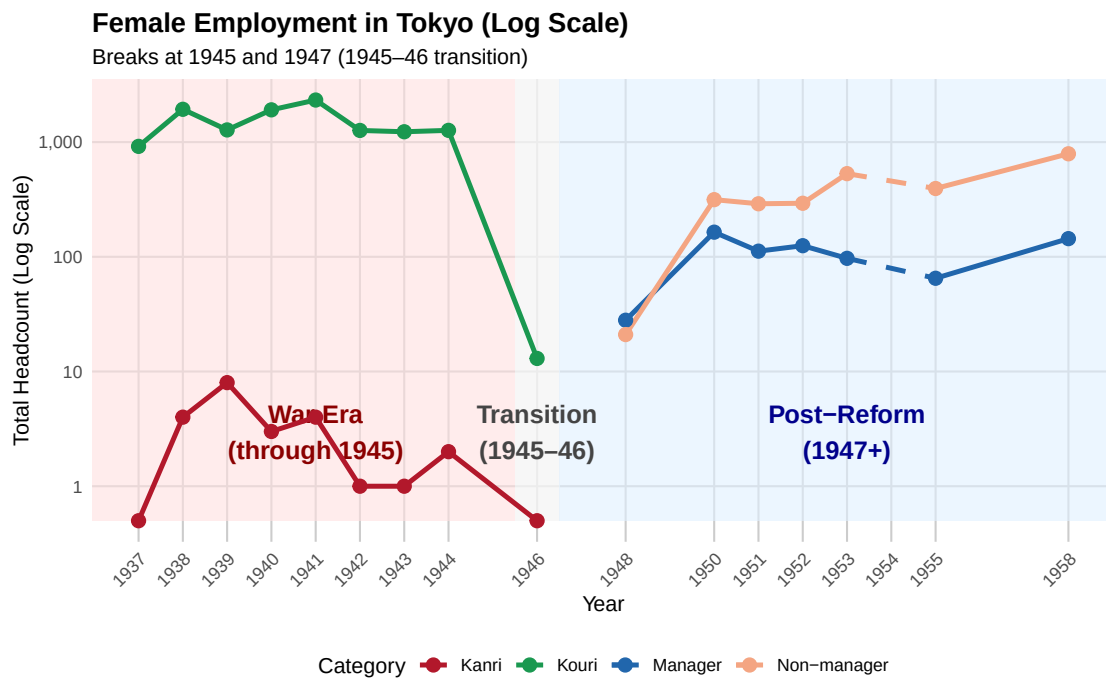
*Clustered (office_id) standard-errors in parentheses**Signif. Codes: ***: 0.01, **: 0.05, *: 0.1***Panel B: Reduced Form (Dep. var: Share retained/promoted post-war)**

Model:	Ret. All (1)	Prom. All (2)	Ret. Male (3)	Prom. Male (4)	Ret. Female (5)	Prom. Female (6)
<i>Variables</i>						
No. males drafted	-0.0003 (0.0012)	-0.0009 (0.0006)	-0.0005 (0.0013)	-0.0010 (0.0007)	0.0067 (0.0042)	0.0014 (0.0011)
<i>Fixed-effects</i>						
year_num	Yes	Yes	Yes	Yes	Yes	Yes
norm_kyoku	Yes	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>						
Observations	1,318	1,318	1,308	1,308	453	453

*Clustered (office_id) standard-errors in parentheses**Signif. Codes: ***: 0.01, **: 0.05, *: 0.1***Panel C: IV (Dep. var: No. retained/promoted post-war)**

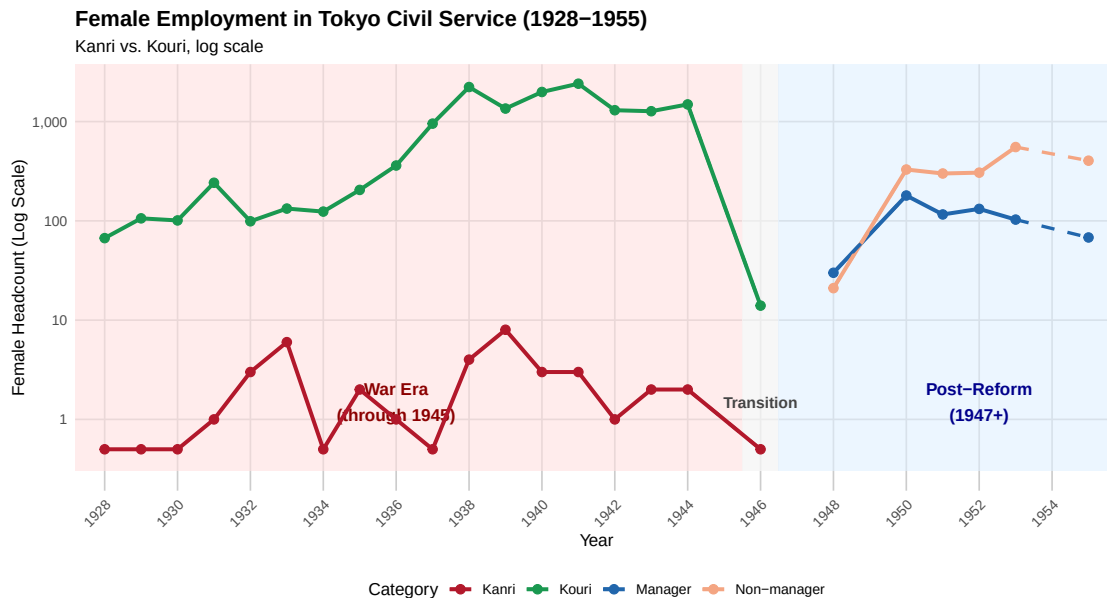
Model:	Ret. All (1)	Prom. All (2)	Ret. Male (3)	Prom. Male (4)	Ret. Female (5)	Prom. Female (6)
<i>Variables</i>						
No. new hires	0.4404 42(0.5216)	-0.6033 (1.368)				
No. male new hires			0.5412 (0.7809)	-0.7856 (2.100)		
No. female new hires (low-rank)					0.1032	0.0067

Figure 1: Female Employment in Tokyo by Rank Category



Note: Headcounts are shown on a log scale; the gap reflects the 1946-1949 transition period.

Figure 2: Women in the Civil Service, 1928–1955



Note: Counts are shown on a log scale. Solid lines indicate observed data; dashed segments connect across gaps in the source records. *Kanri* denotes appointed officials; *Kouri* denotes public employees.

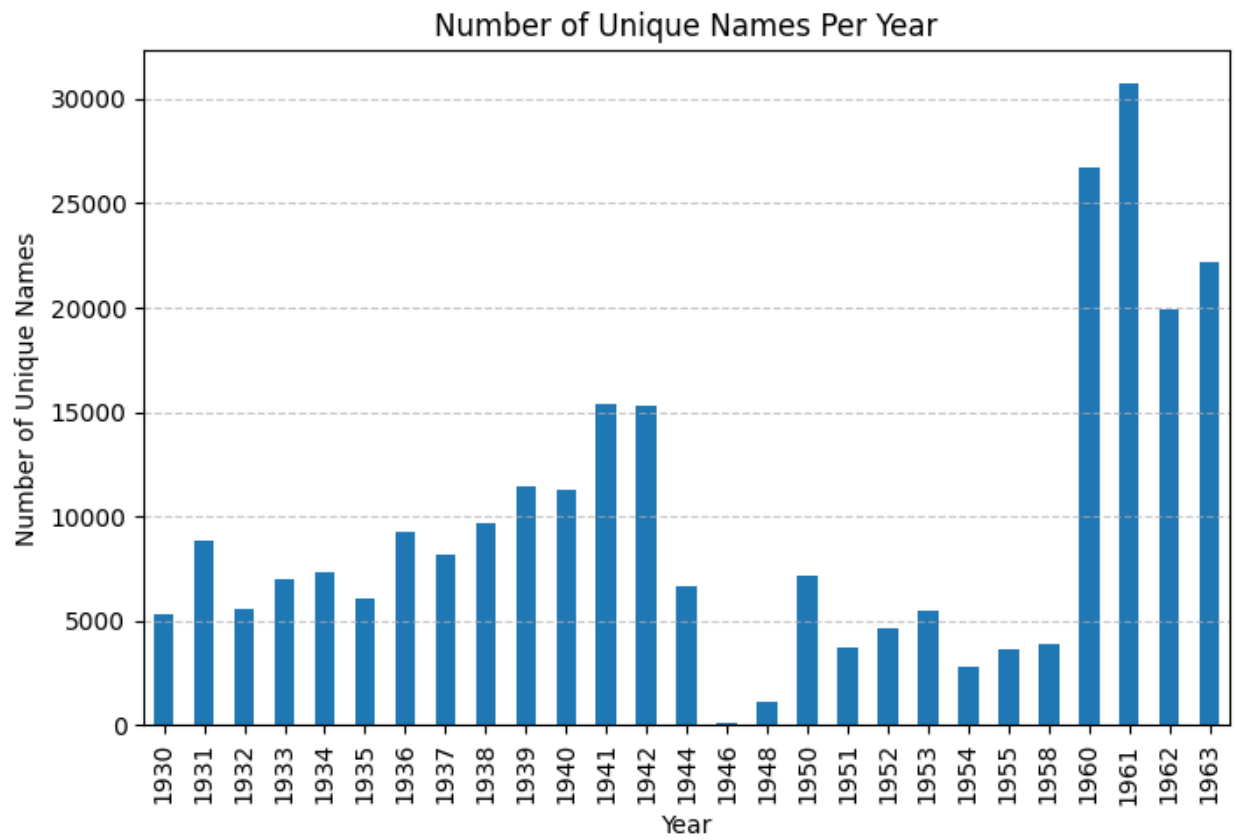
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Figure 3: Number of unique names by year



A Appendix

Table 19: Wartime Female Exposure and Postwar Female Inclusion in Kakari

	Has Female (Binary)			Number of Females		
	(1) Kakaricho (Max)	(2) Kakaricho (Mean)	(3) Kacho (Max)	(4) Kakaricho (Max)	(5) Kakaricho (Mean)	(6) Kacho (Max)
Kakaricho Exposure (Max)	0.418*** (0.092)			1.165** (0.374)		
Kakaricho Exposure (Mean)		0.715*** (0.141)			1.810** (0.563)	
Kacho Exposure (Max)			0.242 (0.200)			−0.485 (0.743)
Engineer Share (Ka)	0.015 (0.040)	0.019 (0.040)	0.309*** (0.091)	−0.381** (0.136)	−0.374** (0.137)	−0.449 (0.637)
No. Kakaricho (Ka)	0.018*** (0.003)	0.018*** (0.003)	0.016*** (0.004)	−0.036 (0.027)	−0.036 (0.027)	−0.096* (0.045)
Kakari Size	0.006*** (0.001)	0.006*** (0.001)	0.005*** (0.001)	0.061*** (0.014)	0.061*** (0.014)	0.076*** (0.014)
Kyoku FE	Yes	Yes	Yes	Yes	Yes	Yes
Clustering	Office	Office	Office	Office	Office	Office
Observations	2,324	2,324	613	2,324	2,324	613
R^2 (within)	0.174	0.177	0.247	0.502	0.502	0.562

Notes: Columns 1–3 estimate a linear probability model $1[\text{Female}_{j,t} > 0] = \beta \text{Exposure}_{m(j)} + X'_{j,t}\gamma + \iota_k + \varepsilon_{j,t}$; columns 4–6 estimate $N^f_{j,t} = \beta \text{Exposure}_{m(j)} + X'_{j,t}\gamma + \iota_k + \varepsilon_{j,t}$ by OLS. The unit of observation is kakari $\times$ year (postwar, 1947–). Exposure is the share of females in the manager $m(j)$'s wartime (1937–1945) position $\times$ kakari $\times$ year cell, aggregated as maximum (cols. 1, 3, 4, 6) or mean (cols. 2, 5) across wartime years. Kakaricho = subsection chief; Kacho = section chief. Controls include engineer share, number of kakaricho, and kakari size. Standard errors clustered at office level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Panel A: Second Stage (Dep. var.: Number of Females)

Table 20: Direct Draft Shock: OLS vs. IV

	OLS (Uncond) (1)	IV (Uncond) (2)	OLS (Cond) (3)	IV (Cond) (4)
Exposure (uncond)	1.730*** (0.133)	5.033* (2.771)		
Exposure (cond)			1.266*** (0.241)	3.846 (16.56)
Engineer share	0.047** (0.019)	0.029 (0.026)	−0.043 (0.075)	0.025 (0.447)
No. Kakari-cho	−0.001 (0.009)	−0.005 (0.009)	0.006 (0.009)	0.003 (0.022)
Kakari size	0.030*** (0.005)	0.031*** (0.004)	0.027*** (0.006)	0.028*** (0.008)
Year FE	Yes	Yes	Yes	Yes
Kyoku FE	Yes	Yes	Yes	Yes
Observations	6,662	6,662	2,324	2,324
R^2	0.275	0.207	0.341	0.298
F-test (1st stage)		3.187		0.074

Panel B: First Stage (Dep. var.: Exposure)

	OLS (Uncond) (1)	IV (Uncond) (2)	OLS (Cond) (3)	IV (Cond) (4)
Draft shock (uncond)		−0.029*** (0.010)		
Draft shock (cond)				−0.005 (0.019)
Engineer share		0.003 (0.008)		−0.021 (0.013)
No. Kakari-cho		0.001*** (0.000)		0.001* (0.001)
Kakari size		−0.000*** (0.000)		−0.000 (0.000)
Year FE		Yes		Yes
Kyoku FE		Yes		Yes
Position FE		Yes		Yes
Observations		6,661		2,324
R^2		0.079		0.064

Notes: Panel A reports estimates of $Y_{j,t} = \beta \text{Exposure}_{j,t} + X'_{j,t}\gamma + \iota_k + \lambda_t + \varepsilon_{j,t}$, where Y is the number of females in kakari j in postwar year t (1947–) and Exposure is the mean share of females in the position $\times$ kakari $\times$ year cell during wartime (1937–1945). Columns 1–2 are unconditional OLS and IV; columns 3–4 are conditional OLS and IV. IV specifications instrument Exposure with the wartime draft shock (share of males drafted); Panel B reports the corresponding first stage in columns 2 and 4. “Cond” conditions on kakaricho (subsection chief) wartime offices; “Uncond” uses all wartime survivors. Controls include engineer share, number of kakaricho, and kakari size. Standard errors clustered at the office level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 21: New Hiring in Response to Adjacent Military Drafting

Model:	Own Drafts (1)	Adj: Diff Kakari (2)	Adj: Diff Occ (3)
<i>Variables</i>			
No. drafted workers	0.6736*** (0.1172)		
Adj. drafts (diff kakari, same occ)		0.0746 (0.0718)	
Adj. drafts (diff occ, same kakari)			0.0062 (0.0942)
<i>Fixed-effects</i>			
year_num	Yes	Yes	Yes
Ka	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes
<i>Fit statistics</i>			
Observations	7,409	7,409	7,409

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Unit of observation: position $\times$ kakari $\times$ year (1938–1945). OLS regressions. Dependent variable: number of new hires. Column (1) uses own-position drafts as treatment. Column (2) uses drafts at different kakari within the same occupation. Column (3) uses drafts at different occupation within the same kakari. All specifications include year, ka, and position fixed effects, and control for log cumulative male baseline. Standard errors clustered at the office level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 22: Worker Reallocation in Response to Military Drafting (1944 Only)

Dependent Variables:	n_transfers_in Transfers In (1)	n_transfers_same_ka Same Ka Transfers (2)	n_transfers_diff_ka Diff Ka Transfers (3)	n_transfers_dist2 Same Kyoku Transfers (4)	n_transfers_diff_ka Diff Ka Transfers (5)
<i>Variables</i>					
Own drafts (male)	0.0367*** (0.0038)	0.0024 (0.0119)	0.0368*** (0.0038)	0.0274** (0.0112)	0.0368*** (0.0038)
log(Male baseline + 1)	-0.0927 (0.1038)		-0.0920 (0.1035)	-8.800*** (0.4304)	-0.0920 (0.1035)
Adj. drafts (diff kakari, same occ)					
Adj. drafts (diff occ, same kakari)					
<i>Fixed-effects</i>					
kyoku	Yes	Yes	Yes	Yes	Yes
pos_norm	Yes	Yes	Yes	Yes	Yes
<i>Fit statistics</i>					
Observations	802	60	802	184	802

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Poisson regressions restricted to 1944. Unit of observation: position $\times$ kakari. Column specifications mirror Table 3. All specifications include office fixed-effects. Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 23: Gender Composition of Hiring and Workforce (1944 Only)

Dependent Variables:	n_new_female	n_new_male	n_new_female	n_new_male	female_share_all		
	Female	Male	Female	Male	Fem. Share	Fem. Share	
	Hires	Hires	Hires	Hires	(All, R1)	(All, R2)	
Model:	(1)	(2)	(3)	(4)	(5)	(6)	
	Poisson	Poisson	Poisson	Poisson	OLS	OLS	
<i>Variables</i>							
Own drafts (male)	0.0382** (0.0179)	0.0353*** (0.0066)	0.0374** (0.0187)	0.0338*** (0.0067)			
log(Male baseline + 1)	-0.4911*** (0.1375)	0.2378*** (0.0731)	-0.5366*** (0.1876)	0.2069** (0.0836)			
Own drafts × Rank 1			0.0181 (0.0499)	0.0530*** (0.0198)			
Any drafted (=1)					-0.3113 (0.2051)	-0.0115 (0.0142)	
Male baseline					-7.31×10^{-5} (7.13×10^{-5})	-0.0009* (0.0005)	-1 (2)
<i>Fixed-effects</i>							
ka	Yes	Yes	Yes	Yes	Yes	Yes	
pos_norm	Yes	Yes	Yes	Yes	Yes	Yes	
<i>Fit statistics</i>							
Observations	232	370	232	370	45	1,246	

Clustered (office_id) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Restricted to 1944. Column specifications mirror Table 9. Poisson regressions for hire counts; OLS for female shares. All position fixed effects. Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 24: Postwar Retention of 1944 Colleagues (1946–1955)

Dependent Variable:	tenure_postwar Retention	
Model:	(1)	(2)
<i>Variables</i>		
Draft share	-0.1288 (0.2345)	-0.0419 (0.2399)
Female (=1)		-0.5192*** (0.1307)
Draft share $\times$ Female		-0.3388 (0.4196)
<i>Fixed-effects</i>		
ka_id_1944	Yes	Yes
pos_norm_1944	Yes	Yes
<i>Fit statistics</i>		
Observations	3,398	3,398
R ²	0.08863	0.09526

Clustered (office_id_1944) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: OLS regressions. Sample: non-drafted workers observed in 1944. Dependent variable is count of postwar years (1946–1955) appearing in civil service. Column (2) includes gender interaction. All specifications include ka and position fixed effects from 1944 and control for log cumulative male stock. Standard errors clustered at the office level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.